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Literature survey on Industrial Automation Load Management System

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***Abstract*— Industrial facilities frequently operate multiple electrical loads such as motors, pumps, compressors, HVAC equipment, and conveyor systems. Manual switching and inadequate coordination of such loads can introduce response delays, unsafe operator exposure, equipment stress, and inefficient power utilization. This paper presents an Industrial Automation Load Management System developed as a relay–contactor based industrial automation prototype. The proposed architecture separates the low-power control domain from the high-power three-phase load domain. A regulated 5 V control supply drives a four-channel relay module, which energizes or de-energizes a three-phase industrial contactor. The contactor performs the actual switching of high-current loads, thereby avoiding direct high-current switching through the low-power relay electronics. The prototype incorporates an AC-to-DC power-conditioning path consisting of a bridge rectifier, filtering stage, and LM117 voltage regulation, together with a 5 V switched-mode power supply for the control circuit. The design emphasizes electrical isolation, modularity, reduced manual intervention, reliable switching, and expandability. The paper describes the system architecture, hardware, operating sequence, applications, advantages, limitations, and a roadmap toward PLC, SCADA, IoT, cloud monitoring, energy analytics, predictive maintenance, and AI-based load prediction. Since the supplied project documentation does not report quantitative energy-saving measurements or laboratory performance datasets, the paper limits its claims to the demonstrated architecture and functional objectives rather than asserting unmeasured numerical improvements.**

***Keywords*-- Industrial automation, load management, relay, contactor, three-phase load, electrical isolation, energy management, Industry 4.0**

I. INTRODUCTION

Industrial automation is the application of control systems, electronic circuits, and intelligent devices to operate industrial processes with minimal human intervention. In an industrial environment, several electrical loads may operate simultaneously, making safe switching and coordinated power utilization important for productivity, equipment protection, and operator safety. The project documentation identifies overloading, manual switching delays, high maintenance cost, improper switching, electrical fault and arc hazards, energy wastage, and poor power-distribution management as practical concerns motivating the proposed system.

The present work addresses the control layer of this problem with a practical relay contactor architecture. Instead of connecting a low-power relay directly to a heavy industrial load, the relay is used to control the coil of a three-phase contactor. The contactor then performs the high-current switching. This creates a modular separation between control electronics and the industrial power path Industrial automation involves the use of control systems, electronic

circuits, sensors, and intelligent devices to operate industrial processes with minimum human intervention. Modern industrial plants contain several electrical loads that may need to be switched according to production requirements. Effective management of these loads is important for improving safety, energy efficiency, productivity, and equipment reliability. The project PPT identifies overloading, manual switching delays, improper switching, electrical faults, arc hazards, energy wastage, and poor power distribution as major industrial problems.

Traditional industrial load control often depends on manual operation or separate control mechanisms. Manual monitoring can introduce response delays and may result in inappropriate switching sequences. When high-power loads are switched incorrectly, motors, compressors, pumps, and other expensive equipment can experience electrical and mechanical stress. To address these challenges, this project proposes an Industrial Automation Load Management System based on a relay contactor switching architecture. The system uses a low-voltage control circuit to operate a relay module, while the relay controls the coil of a three-phase contactor. The contactor then connects or disconnects the high-power industrial load. This architecture allows low-power electronics to control heavy electrical equipment while maintaining isolation between the control and load sides.

II. LITERATURE REVIEW ON LOADSENSE

In This Review We Survey 10 Paper based on That.

In The 1st Paper Industrial Power Load Scheduling Considering Demand Response

Cui and Zhou [1] studied industrial power load scheduling considering demand response in smart-grid environments. Their research examined methods for scheduling industrial electrical loads to reduce electricity costs, improve grid reliability, and balance power demand. The study discussed single-objective and multi-objective optimization techniques while considering production constraints, economic benefits, and environmental factors. The findings highlight the importance of intelligent load scheduling in improving energy efficiency without interrupting essential industrial operations. However, implementing these optimization methods in real-time industrial environments can be challenging because of changing production requirements and electricity prices. This study provides a useful reference for developing an Industrial Automation Load Management System that prioritizes electrical loads and controls power consumption according to operating conditions.

In This 2nd Paper Industrial Energy Management System: Design of a Conceptual Framework Using IoT and Big Data

Ullah et al. [2] proposed a conceptual framework for an Industrial Energy Management System using the Internet of Things (IoT) and big data technologies. Their study reviewed existing industrial energy management approaches and identified the major requirements for collecting, processing, analyzing, and presenting energy-related data. The proposed framework integrates data acquisition and analytics to support better energy-related decision-making in industrial environments. The study emphasizes the potential of connected sensors and data-driven monitoring to improve energy efficiency and operational visibility. However, implementing such a framework requires suitable communication infrastructure, reliable data collection, and integration with existing industrial equipment. This research is relevant to an Industrial Automation Load

Management System because it supports the use of sensors, monitoring units, and automated control for efficient industrial energy utilization.

In This 3rd Paper Planning and Scheduling for Industrial Demand-Side Management: State of the Art, Opportunities and Challenges

S. Chen, F. Gong, [3] reviewed industrial demand-side management techniques and the challenges associated with scheduling industrial electricity consumption. Their study examined how industrial loads can be adjusted according to electricity demand, grid conditions, and the availability of renewable energy. The research highlighted the potential of automated demand response in industries with flexible production processes and controllable equipment. It also discussed opportunities arising from the integration of industrial Internet technologies and energy management systems. However, load scheduling must consider production continuity, equipment operating limits, and the complexity of coordinating multiple industrial loads. This study provides a theoretical foundation for designing an automated load management system that assigns priorities to electrical equipment and reduces unnecessary power consumption during peak-demand periods.

In This 4th Paper Two-Stage Discrete-Continuous Multi-Objective Load Optimization for Industrial Demand Response

The authors of [4] investigated a two-stage discrete-continuous multi-objective optimization method for industrial demand response. The proposed approach combines load-shifting decisions with continuous optimization of controllable loads, taking into account electricity prices and the operational requirements of industrial consumers. The study demonstrated the application of the method to an industrial building model using real data. Its main advantage is the ability to coordinate load scheduling while considering both economic benefits and consumer requirements. However, optimization-based scheduling may require accurate load information and careful integration with real-time industrial control systems. This research is useful for an Industrial Automation Load Management System because it demonstrates how loads can be scheduled or shifted intelligently to manage peak demand while maintaining essential operations.

In This 5th Paper IoT-Based Demand-Side Energy Management Through Automated Load Control

Tweve et al. [5] presented an IoT-based demand-side energy management framework for industrial solar microgrids. Their system combines real-time electrical monitoring, automated power-factor correction, and priority-based control of non-critical loads. The authors evaluated the approach using simulation, a physical prototype, and industrial load data. Their reported results showed that automated shedding of non-critical loads reduced peak-period real power demand, demonstrating the potential of sensor-based monitoring and automatic control to improve electrical performance. However, the study also identifies the need for further evaluation across different load profiles and operating environments. This research is particularly relevant to an Industrial Automation Load Management System because it demonstrates how real-time monitoring and automatic switching of non-essential loads can help manage peak demand and improve energy utilization.

In This 6th Paper Load Management Using PLC and SCADA

Logic Controller (PLC) [6] and Supervisory Control and Data Acquisition (SCADA). The proposed system manages multiple electrical loads by assigning priority levels and regulating their operating durations through timer-based PLC programming. Lower-priority loads are automatically disconnected when required to maintain operation within a predefined energy limit, while the SCADA interface enables real-time monitoring of load status and system performance. The integration of PLC and SCADA aims to reduce manual intervention, improve load distribution, and enhance the reliability of industrial energy management. However, the available publication abstract does not report quantitative experimental results demonstrating energy savings or peak-demand reduction. The study provides a useful reference for developing an industrial automation load management system with priority-based switching and centralized monitoring.

In This 7th Paper Automatic Priority Based Load Shedding Management

Maity et al. [7] presented an automatic priority-based load-shedding scheme for an industrial cogeneration power plant in Taloja MIDC, Maharashtra. The study addresses the problem of generator overloading when the main grid supply trips and the remaining alternator is required to supply the connected industrial loads. The proposed scheme uses predefined load-shedding steps, appropriate load-block sizes, and frequency-relay settings to disconnect selected loads and prevent system instability. The authors also emphasize controlled load restoration, in which loads are reconnected in smaller blocks with suitable time delays after the system frequency has returned to its normal operating condition. The study highlights the importance of coordinating load priorities, generation capacity, and restoration sequences in industrial power systems. However, further assessment of the implementation details and quantitative performance results is necessary to establish the effectiveness of the scheme under different operating conditions. This research provides a useful foundation for developing an Industrial Automation Load Management System that supports automatic load disconnection, priority-based control, and safe restoration following power-supply disturbances.

In This 8th Paper An Intelligent Load Priority List–Based Integrated Energy Management System in Microgrids

Alahmed and Al-Muhaini [8] proposed an intelligent load priority list-based integrated energy management system for microgrids using a neuro-fuzzy demand-side management approach. Unlike conventional methods that rely on a fixed load-priority list, the proposed system dynamically ranks load categories according to factors such as load importance, time of day, system reliability, and available energy supply. The methodology was implemented on a low-voltage microgrid testbed incorporating hybrid renewable generation, energy storage, and multiple load categories, with MATLAB and Simulink used for simulation. The study investigated energy-balance-based demand management, priority-based load curtailment, peak shaving, and load shifting. The results demonstrated that adaptive load prioritization could improve the utilization of local generation, reduce dependence on expensive grid electricity

during high-tariff periods, and minimize curtailment of essential loads. However, practical implementation in industrial environments requires appropriate measurement systems, reliable control infrastructure, and consideration of production constraints. This research provides a valuable foundation for developing an Industrial Automation Load Management System that incorporates intelligent load prioritization and demand-based power control.

In This 9th Paper Optimal Load Scheduling of Demand Response in Heavy Industrial Facility Based on Hierarchical Optimization

Yu et al. [9] proposed a hierarchical optimization-based load-scheduling method for demand response in heavy industrial facilities, using a steel manufacturing enterprise as a case study. The proposed methodology divides industrial equipment into functional zones according to production processes and further classifies equipment based on its operational importance and demand-response capability. A multi-objective optimization model was developed to account for electricity-price variations, scheduling pressure, equipment response costs, and operational constraints. The model was implemented using MATLAB 2024a and solved using CPLEX over a 24-hour scheduling horizon. The reported results indicate that zonal and hierarchical scheduling can improve load coordination, utilize flexible demand-response resources, and reduce operational costs while protecting essential production equipment. However, the approach requires accurate equipment data, appropriate parameter tuning, and sufficient computational resources for large-scale or near-real-time applications. This study provides a useful reference for developing an Industrial Automation Load Management System that incorporates equipment prioritization, coordinated load scheduling, and demand-based power control.

In This 10th Paper A Real-Time Decision Model for Industrial Load Management in a Smart Grid

Yu, Lu, and Hong [10] proposed a real-time decision model for industrial load management in a smart-grid environment. Their research focused on controlling industrial electricity consumption in response to changing real-time electricity prices while considering the operational dependencies between consecutive manufacturing tasks. The proposed approach uses robust optimization to manage uncertainties in future electricity prices and make timely load-control decisions. The researchers evaluated the model using a steel powder manufacturing process, and the simulation results demonstrated the effectiveness of the proposed scheduling approach. The main advantage of this method is that it considers both current load-control decisions and future production requirements, helping to balance electricity costs and manufacturing continuity. However, practical implementation requires reliable electricity-price information, accurate production data, and integration with industrial monitoring and control systems. This research provides a valuable reference for developing an Industrial Automation Load Management System that monitors electrical loads, prioritizes equipment, and controls power consumption according to operating conditions while maintaining essential industrial operations.

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III. COMPARATIVE ANALYSIS OF SYSTEM

Comparative Analysis of Relevant Research Papers

(Industrial Automation Load Management System)

Ref. / Author (Year)	Methodology	Key Findings	Research Gap / Relevance to Proposed System
[1] Cui and Zhou (2018)	• Demand-response-based load scheduling and optimization.	• Balances demand and cost through optimized scheduling.	• Needs accurate operating data and production constraints; supports coordinated load control.
[2] Ullah et al. (2022)	• IoT and big-data-based energy management framework.	• Enables energy monitoring and data-driven decision making.	• Requires communication infrastructure; supports future IoT monitoring.
[3] Chen et al. (2021)	• Review of industrial demand-side management.	• Supports flexible load scheduling and automated response.	• Production continuity and coordination of multiple loads remain challenging; informs priority-based management.
[4] Abdulaal et al. (2017)	• Two-stage multi-objective optimization.	• Considers electricity prices and industrial operating constraints.	• Requires accurate load information and control systems; supports demand-aware scheduling.
[5] Tweve et al. (2026)	• IoT monitoring and automated priority-based load control.	• Shows potential for automated non-critical load shedding.	• Further validation across different load profiles is needed; supports automated switching.
[6] Kindarle et al. (2026)	• PLC, SCADA, and timer-based priority control.	• Enables centralized monitoring and automatic load control.	• Reviewed abstract does not report quantitative performance improvements; possible future extension using PLC and SCADA.
[7] Maity et al. (2023)	• Frequency-relay-based priority load shedding.	• Supports staged disconnection and controlled restoration.	• Further quantitative evaluation is required; supports priority-based switching during disturbances.
[8] Alahmed and Al-Muhaini (2020)	• Neuro-fuzzy load prioritization.	• Supports adaptive prioritization and demand management.	• Requires suitable measurements and reliable control infrastructure; basis for intelligent load-priority logic.
[9] Yu et al. (2026)	• Hierarchical load-scheduling optimization.	• Coordinates loads considering equipment importance and constraints.	• Needs accurate equipment data and computational resources; supports coordinated scheduling.
[10] Yu, Lu, and Hong (2016)	• Robust real-time load-management decision model.	• Considers electricity prices and manufacturing-task dependencies.	• Requires reliable price, production, and monitoring data; supports future real-time load-management capabilities.

Table 1. Comparative analysis of selected research works relevant of Load sense

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IV. BLOCK DIAGRAM OF PROPOSED SYSTEM

The block diagram of proposed system shown below illustrating the overall architecture.

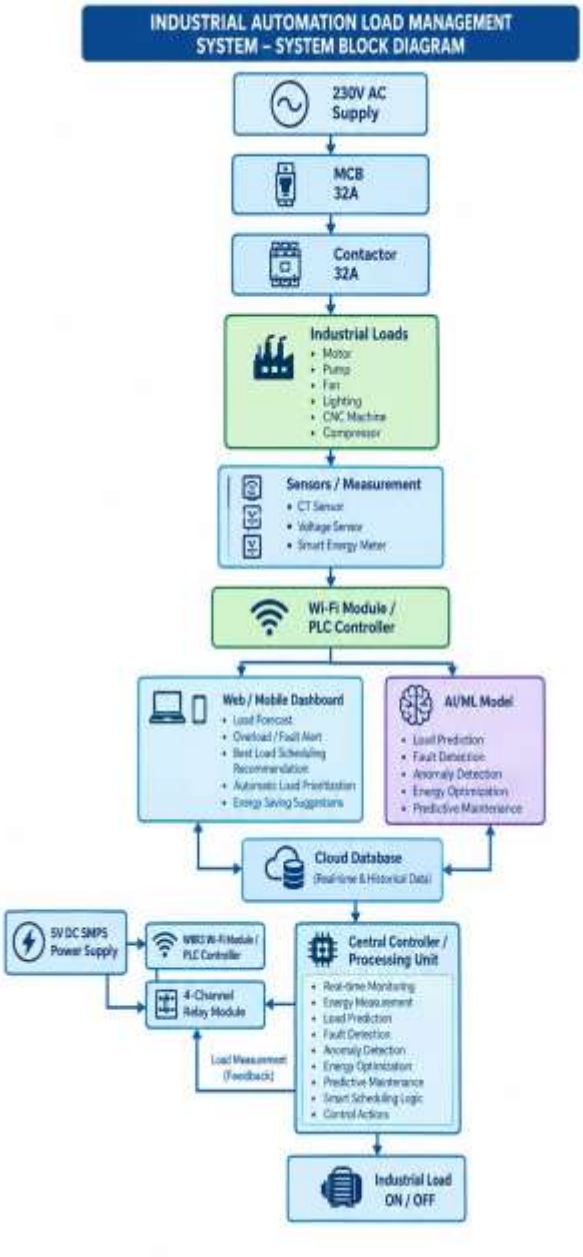


Figure 1 Proposed Load Sense system block diagram

V. BELOW IS THE FLOW OF PROPOSED SYSTEM

The flow diagram of the proposed system is shown below illustrating the overall architecture

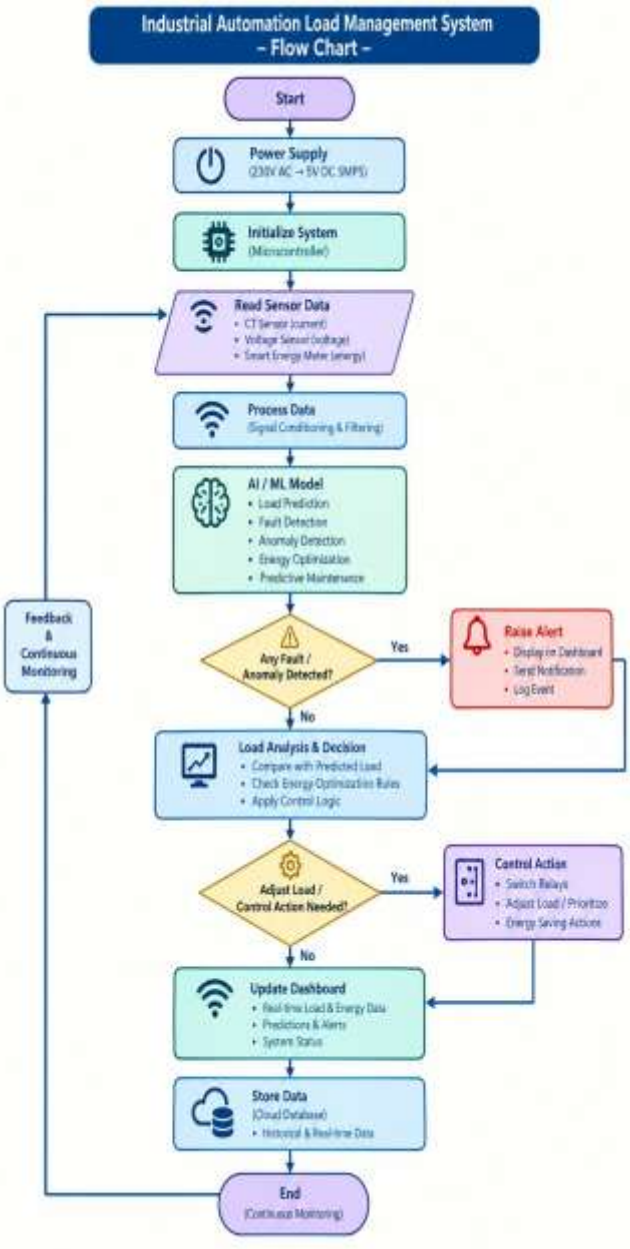


Figure2: Flowchart of the proposed Load Sense system.

The proposed Industrial Automation Load Management System will control and manage multiple industrial electrical loads using a relay contactor switching mechanism. The low-voltage control circuit will activate the relay module, which energizes the contactor to switch three-phase industrial loads such as motors, pumps, and conveyors safely. energizes the contactor to switch three-phase industrial loads such as motors, pumps, and conveyors safely. The system will use predefined control logic and load priorities to switch electrical loads ON or OFF, reduce unnecessary power consumption, and minimize manual intervention. Electrical isolation between the control and power circuits will help improve operational safety. The proposed system will serve as an educational and research prototype for automated industrial load management. It can be further enhanced with real-time power monitoring, PLC, SCADA, and intelligent load scheduling to improve energy efficiency and industrial automation.

VI. CONCLUSION

The literature survey shows that industrial load management systems play an important role in improving energy efficiency, reducing electricity costs, balancing power demand, and ensuring reliable industrial operations. The reviewed studies highlight the benefits of demand response, IoT-based monitoring, PLC and SCADA integration, priority-based load shedding, and intelligent load scheduling. However, challenges remain in implementing real-time control, integrating existing industrial equipment, and maintaining production continuity. Based on these studies, the proposed Industrial Automation Load Management System aims to automate the switching of industrial electrical loads using a relay contactor mechanism and predefined control logic. The system will prioritize loads, reduce unnecessary power consumption, and minimize manual intervention. In the future, it can be enhanced with real-time power monitoring, PLC, SCADA, and intelligent scheduling techniques to achieve more efficient, reliable, and automated industrial energy management.

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